STAND-HF Functional Acceptance

19. Functional Acceptance Criteria

One section per screen or form. For each one: who uses it, the fields, the validation, and the checks the client signs off on.

Rules used throughout - **Hard error** blocks saving as complete and blocks “mark complete”. Drafts can be saved with hard errors. - **Soft warning** is shown as an amber notice. The user can continue. - All numeric ranges are **placeholders** until the client confirms them (Q-01). Marked TBC. - Sources: `[BRIEF sN]` developer brief slide, `[FSD 2.x]` FSD section, `[FR §n]` functional requirements, `[06]` 06-validation-rules.md.

A. Login

Who: all three roles. **Source:** `[FSD 2.2]`, `[BRIEF s7]`.

Field	Required	Rule
Username	Yes	Not empty
Password	Yes	Not empty

Acceptance

1. Correct username and password open the home screen for that user’s role.
2. Wrong username or password shows one message: “Username or password is incorrect”. It does not say which one was wrong.
3. Empty username or password is blocked with a message next to the empty field. No request is sent.
4. After N failed attempts (N TBC) the account is locked for a set time. The locked message is clear. (Lockout is not yet built in the prototype.)
5. A deactivated user cannot log in.
6. Idle session ends after a set time (TBC). The user is sent back to login with a message.
7. Logout ends the session. Going back in the browser does not show data.
8. First login after a password reset asks for a new password.
9. Password must meet the policy (exact rules TBC). A weak password is rejected with the reason shown.

Not in the pilot: SSO and MFA (Q-02).

B. Change password

Who: any logged-in user, for their own password.

Field	Required	Rule
Current password	Yes	Must match
New password	Yes	Meets policy (TBC)
Confirm new password	Yes	Must match new password

Acceptance

1. Wrong current password is rejected.
2. Mismatch between new and confirm is rejected.
3. Password used before cannot be reused (count TBC).
4. Success returns the user to home with a confirmation message.

C. Super Admin: reset a user's password

Who: Super Admin only.

Acceptance

1. Reset generates a temporary password, shown once.
2. The user must change it at next login.
3. The reset is recorded in the audit trail.

D. Users (create, edit, deactivate)

Who: Super Admin creates all users. Central team adds Site Investigators only (see section E). **Source:** `[FSD 2.8]`, `[BRIEF s7]`.

Field	Required	Rule
Username	Yes	Unique
Role	Yes	One of the three roles
Sites	Yes for Site Investigator	One or more active sites. Not needed for Super Admin or Central team
Active	Yes	On or off

Acceptance

1. Creating a user with a duplicate username is rejected.
2. A Site Investigator with no site assigned is rejected.
3. Deactivating a user blocks login. The user is not deleted, and their past records stay.
4. Super Admin cannot change its own role or deactivate its own account.
5. At least one active Super Admin must remain. Deactivating the last one is rejected.
6. Every create, edit and deactivate is recorded in the audit trail.

E. Central team: add site investigators

Who: Central team. **Source:** `[BRIEF s7]`, `[03-roles-access.md]`.

Acceptance

1. The user list shows Site Investigators only. Super Admin and Central team accounts are not shown.
2. Central team can add a Site Investigator, assign sites, and switch access on or off.
3. Central team cannot create or edit Super Admin or Central team accounts. The attempt is rejected.

F. Roles and permissions

Who: Super Admin edits the matrix. Central team and Site Investigator rows can be changed by Super Admin.
Source: `[03-roles-access.md]`.

Permission	Site Investigator (default)	Central team (default)
See patients at every site	No	Yes
Add patients	Yes	No
Enter and edit visit data	Yes	No
Mark visits complete or reopen	Yes	Yes
Export CSV	No	Yes
Add site investigators and assign sites	No	Yes
Manage users	No	No
Manage sites	No	No

Acceptance

1. Super Admin’s row is fixed and cannot be edited.
2. Changing a permission takes effect on the next request.
3. Each change is recorded in the audit trail.

G. Sites

Who: Super Admin. **Source:** `[FSD 2.8]`, `[05-data-model.md]`.

Field	Required	Rule
Site code	Yes	Unique. Used in Study ID (for example RIC, LHR)
Site name	Yes	Not empty

Field	Required	Rule
Active	Yes	On or off

Acceptance

1. Duplicate site code is rejected.
2. Deactivating a site keeps its patients. Super Admin can still read them. Site Investigators assigned only to that site lose access.
3. Sites are deactivated, never deleted.

H. Home screen (per role)

Who: all roles. Each sees different sections. **Source:** `[03-roles-access.md]`, `[07-ui-design.md]`.

Section	Super Admin	Central team	Site Investigator
Study statistics	All sites	All sites	Own sites
Patient list with Baseline, 3M, 6M status	All	All	Own patients
Overdue visits	All	All	Own patients
Users	All	Investigators only	Hidden
Sites	All	View only	Hidden
Permissions	Yes	Hidden	Hidden
Export	Yes (TBC)	Yes	Hidden
Add patient button	No	No	Yes

Acceptance

1. Each role sees only the sections in its column.
2. Overdue visit = due window has passed and the visit is not complete.
3. Numbers on the home screen match a manual count on test data.

I. Patient registration (add patient)

Who: Site Investigator (own site). Central team and Super Admin cannot add patients. **Source:** `[FSD 2.3]`, `[BRIEF s4]`, `[06]`.

Field	Required	Rule	Type
Study ID	Auto	Generated from site code plus sequence. Read only	-

Field	Required	Rule	Type
Site	Yes	Only the user's assigned sites	Hard
Patient name	Yes	Not empty	Hard. New from UI change
Contact number	Yes	Format TBC (Q-19)	Hard. New from UI change
Investigator name	Yes	Not empty	Hard
Consent date	Yes	Not empty	Hard
Age	Yes	18 or more (TBC)	Hard
Sex	Yes	Male or Female	Hard
HF etiology	Yes	Ischemic, Non-ischemic or Other	Hard
Etiology other (text)	If "Other"	Not empty	Hard

Acceptance

1. Saving generates a unique Study ID, for example `RIC-0001`. It shows on screen after saving.
2. A missing required field is blocked, and the field name is shown next to it.
3. A Site Investigator cannot pick a site they are not assigned to.
4. Age below 18 is blocked (TBC).
5. The patient appears in the site's list straight away.
6. Editing a saved patient needs a reason. The change is recorded in the audit trail.
7. Patient name and contact number are required. Saving without either is blocked, and the missing field is named.

J. Patient list

Who: all roles, scoped by site. **Source:** `[FSD 2.3]`.

Acceptance

1. Site Investigator sees only own-site patients.
2. Central team and Super Admin see all sites.
3. Each row shows Study ID, site, Baseline status, 3M status, 6M status and next due date.
4. Search by Study ID works.

K. Baseline visit

Who: Site Investigator. Central team can mark complete or reopen. **Source:** `[BRIEF s3]`, `[FSD 2.4]`, `[06]`.

Contains: visit details, vitals and function, LVEF, GDMT section (see section N), prescription upload (see section R), and no events. Baseline records prior history only (TBC).

Field	Required	Rule	Type
Visit date	Yes	Not before consent date	Hard
Weight (kg)	Yes	Placeholder range (TBC)	Soft warning outside range
Heart rate (bpm)	Yes	Placeholder 30 to 220	Soft warning outside range
Systolic BP (mmHg)	Yes	Placeholder 50 to 260	Soft warning outside range
Diastolic BP (mmHg)	Yes	Must be lower than systolic	Hard
NYHA class	Yes	I, II, III or IV	Hard
NYHA IV	If IV	Confirm exclusion criteria	Soft warning
1MSTS repetitions in 60 s	Yes	Placeholder 0 to 60	Soft warning outside range
1MSTS stopped early (reason)	No	Prototype list (TBC)	-
1MSTS repeated	No	Yes or no	-
LVEF (%)	Yes	Placeholder 5 to 80	Hard if missing. Soft if out of range
LVEF above 40%	-	Outside registry inclusion (TBC)	Soft warning
Echo date	No	Within 30 days before enrollment (TBC)	Soft warning

Acceptance

1. Saving a draft works even with hard errors. The draft shows which fields are missing.
2. “Mark complete” lists all missing or invalid fields. It does not complete the visit while any hard error exists.
3. A soft warning shows as an amber notice. Completion is still allowed.
4. Completing the visit locks it. Site users cannot edit a locked visit.
5. Reopening needs a reason. The reopen and the reason are recorded in the audit trail.
6. Every value changed after saving needs a reason. The old and new values are recorded.

L. 3-month visit

Who: Site Investigator. Central team can mark complete or reopen. **Source:** `[BRIEF s3]`, `[FSD 2.4, 2.6]`, `[06]`.

Contains: visit details, vitals and function (no LVEF), GDMT section, events section, prescription upload (see section R).

Field	Required	Rule	Type
Visit date	Yes	Must be after Baseline date	Hard

Field	Required	Rule	Type
Visit window	-	Day 90 ±14 from Baseline (TBC)	Soft warning outside window
Weight, heart rate, BP, NYHA, 1MSTS	As Baseline	As Baseline	As Baseline
LVEF	Not shown	Not captured at 3M	-
Events (see section M)	Yes	Entered at 3M and 6M only	Hard

Acceptance

1. Visit date before Baseline date is blocked.
2. Visit outside the window shows a soft warning.
3. LVEF is not shown and cannot be entered at 3M.
4. Mark complete and lock work as in section K.

M. Clinical events (inside 3-month and 6-month visits)

Who: Site Investigator enters. Central team reads. **Source:** `[BRIEF s5]`, `[FSD 2.6]`, `[06]`.

Field	Required	Rule	Type
HF hospitalization	Yes (yes or no)	Since previous visit	-
Other hospitalization	Yes (yes or no)	Since previous visit	-
Death	Yes (yes or no)	Since previous visit	-
Major intervention or event	No	Free text	-

Acceptance

1. Events cannot be entered at Baseline. Attempting it returns a hard error.
2. Death recorded at 3M or 6M shows a soft warning: confirm no further visits. The warning does not block saving.
3. Whether death blocks further visits is TBC. Until then, no further visits are blocked.

N. GDMT section (inside every visit)

Who: Site Investigator enters. Central team reads. **Source:** `[BRIEF s6]`, `[FSD 2.5]`, `[06]`.

Four cards, one per pillar: ARNI (sacubitril and valsartan), evidence-based beta-blocker, MRA, SGLT2 inhibitor. The ARNI pillar is first. ACE-I and ARB are not ARNI.

Field	Required	Rule	Type
Status	Yes	Started, Continued, Stopped or Not started	Hard
Generic drug	If Started or Continued	Dropdown only, no free text	Hard
Daily dose	If Started or Continued	Not empty	Hard
Start date	If Started or Continued	Not empty	Soft warning
Reason	If Not started or Stopped	At least one reason from the list	Soft warning

Reason list: Hypotension, Bradycardia, Renal dysfunction, Hyperkalemia, Intolerance, Cost or access, Patient preference, Physician decision, Other (free text).

Active score = count of pillars with status Started or Continued (0 to 4). Shown on the visit and on the patient record.

Acceptance

1. Every pillar must have a status before the visit can be marked complete.
2. Started or Continued without a drug or dose is a hard error.
3. Drug dropdown lists only active drugs for that pillar. ARNI list does not include ACE-I or ARB.
4. Active score updates as statuses change, and reads 0 to 4.
5. Not started or Stopped without a reason gives a soft warning.

O. Visit reopen

Who: Central team and Site Investigator (permission per matrix).

Acceptance

1. Reopen needs a written reason.
2. Reopen is recorded in the audit trail with the user and time.
3. After reopen, the visit can be edited and completed again.

P. Audit trail view

Who: Central team and Super Admin. Site Investigators do not see it. **Source:** `[BRIEF s9]`, `[FSD 2.1]`.

Each row: who changed it, what field, old value, new value, date and time, reason.

Acceptance

1. Changes can be viewed per patient and per user.
2. Each correction has a reason. A correction without a reason cannot be saved.
3. Rows cannot be edited or deleted.
4. Site Investigators get “not allowed” if they open the view.

Q. CSV export

Who: Central team (all sites). Site Investigator only own rows (TBC). **Source:** `[FSD 2.7]`, `[10-export.md]`.

Acceptance

1. Output is one row per patient-visit.
2. Columns match `10-export.md`. Column list confirmed with the data manager (EXP-02).
3. Each row shows site, visit status, GDMT pillars, clinical assessments and events.
4. Incomplete visits are excluded (TBC).
5. Each download is recorded in the audit trail.
6. Values are not changed during export.

R. Prescription upload (Baseline, 3M, 6M)

Who: Site Investigator uploads at their own site. Central team and Super Admin can view. **Source:** UI change (new). File limits are defaults until Q-20 is answered.

Field	Required	Rule	Type
File	No	PDF, JPEG or PNG	Hard
File size	-	Default 10 MB per file (TBC)	Hard
Number of files	-	No limit set (TBC)	-

Acceptance

1. The “Upload prescription” button is on the Baseline, 3M and 6M forms.
2. A PDF, JPEG or PNG under the size limit uploads and appears in the visit’s file list.
3. Any other file type is rejected with the allowed types shown.
4. A file over the size limit is rejected with the limit shown.
5. Uploading is not allowed on a locked visit. The user is told to reopen it first.
6. Each upload is recorded in the audit trail with the user, time and file name.
7. Upload does not block mark complete.

S. Patient record (information and files)

Who: all roles, scoped by site. **Source:** UI change (new).

Shows, for each of the three visits: visit information, GDMT score, events, and the uploaded files. Patient name and contact number are shown at the top.

Acceptance

1. All patient details, including patient name and contact number, appear on the record.
2. Each visit lists its files with name, type, size and upload date.
3. A file opens or downloads from the record for users with access to that site.
4. Site Investigators cannot open files from another site. The attempt is refused.
5. Each file view is recorded in the audit trail (TBC, Q-20).

Open client items that affect these forms

Item	Affects
Q-01 numeric ranges	All vitals and LVEF rules (K, L)
Q-02 login method	A
Q-04 edit after sync	I (edit rules)
Q-06 and Q-07 multi-site and role scope	D, F, Q
Q-08 audit scope	P
Q-16 drug list	N
Q-17 visit windows	K, L
Q-18 LVEF cutoff and minimum age	I, K
Q-19 contact number format	I
Q-20 file size, types, count, retention, who can delete	R, S
Q-21 where files are stored (personal data, hosting)	R, S